Q. List out the important food safety standards and regulations – 10 M OR
Q. Expand the term FSSAI, illustrate its functions, importance & type of license issued by them and its role in FBO – 10 M

Answer:

Important food safety standards and regulations:

1. **FSSAI (Food Safety and Standards Authority of India)**
    - Full form: Food Safety and Standards Authority of India.
    - Belongs to the Ministry of Health and Family Welfare, Government of India.
    - Establishes scientific standards for control, inspection, and regulation of production, storage, sale, distribution, and import of food.
    - Ensures safe, healthy, and nutritious food for consumers.
2. **Functions of FSSAI**
    - Framing laws and setting food safety standards.
    - Regulating manufacturing, storage, distribution, sale, and import of food.
    - Accrediting laboratories for food testing.
    - Providing technical assistance and scientific guidance to the Central Government.
    - Conducting surveys on food intake and food pollution.
    - Spreading awareness regarding food quality.
    - Conducting training programmes for food business operators.
    - Ensuring food is free from contaminants and preventing food adulteration.
    - Inspecting quality, authenticity, shelf life, and sensory characteristics of food.
3. **Licensing regulations**
    - FSSAI Basic Registration: For small businesses with annual turnover below Rs.12 lakhs.
    - FSSAI State License: For small and medium businesses with turnover between Rs.12 lakhs and Rs.20 crores.
    - FSSAI Central License: For large businesses with turnover above Rs.20 crores; also required for import or export of food products.
4. **Compliance rules**
    - Food license is compulsory for food business operators.
    - Adhering to food safety and labeling standards.
    - Maintaining proper hygiene and sanitation in processing and handling.
    - Regular inspections, audits, and food testing in accredited laboratories.
    - Proper packaging, labeling, and record keeping during sampling and testing.
5. **Importance of food safety regulations**
    - Protects consumers from foodborne illnesses and contaminated food.
    - Standardizes food quality and ensures transparency in labeling.
    - Builds consumer trust and improves brand reputation.
    - Supports international trade by meeting global safety standards.
6. **Role of FSSAI in FBO (Food Business Operators)**
    - Obtaining an FSSAI license is compulsory for food business operators.
    - Ensures FBOs follow food safety and labeling standards.
    - Requires proper hygiene and sanitation in processing and handling.

    - Conducts inspections, audits, and food testing to ensure compliance.
    - Ensures safe production, storage, and distribution of food to protect consumers.

Q. Explain various biochemical analysis of food – 8 M

Answer:

Biochemical analysis of food:

Biochemical analysis of food involves examining the chemical composition and properties of food items to ensure quality, safety, and nutritional value.

1. Nutritional Analysis

- Purpose: Determines nutritional content of food, including macronutrients (proteins, fats, carbohydrates) and micronutrients (vitamins, minerals).
- Methods:
    - Proximate analysis: Measures moisture, ash, protein, fat, and fiber content.
    - Calorimetry: Assesses the energy content of food.

2. Protein Analysis

- Purpose: Identifies the quantity and quality of proteins in food.
- Methods:
    - Kjeldahl method: Measures total nitrogen to estimate protein content.
    - Biuret test: Detects peptide bonds in proteins.

3. Fat Analysis

- Purpose: Analyzes the type and amount of fats and oils in food.
- Methods:
    - Soxhlet extraction: Extracts fat from food samples for quantification.
    - Gas chromatography (GC): Analyzes fatty acid profiles.

4. Carbohydrate Analysis

- Purpose: Measures different types of carbohydrates, including sugars, starches, and fiber.
- Methods:
    - Anthrone test: Determines total carbohydrates.
    - HPLC (High-Performance Liquid Chromatography): Separates and quantifies individual sugars.

5. Vitamin Analysis

- Purpose: Assesses the presence and concentration of vitamins in food.
- Methods:
    - HPLC: Used for analyzing water-soluble vitamins such as Vitamin C and B vitamins.

    - Spectrophotometry: Measures absorbance related to vitamin concentration.

6. Mineral Analysis

- Purpose: Determines levels of essential minerals in food.
- Methods:
    - Atomic Absorption Spectroscopy (AAS): Measures concentration of metals like iron, calcium, and magnesium.
    - Inductively Coupled Plasma (ICP): Analyzes multiple elements simultaneously.

7. pH Measurement

- Purpose: Evaluates acidity or alkalinity of food, affecting flavor, preservation, and safety.
- Methods:
    - pH meter: Provides accurate readings of food samples.

8. Enzyme Activity Analysis

- Purpose: Measures activity of enzymes affecting food quality and shelf life.
- Methods:
    - ELISA: Detects and quantifies enzyme levels.
    - Spectrophotometric assays: Monitor absorbance changes related to enzyme reactions.

9. Microbial Analysis

- Purpose: Identifies and quantifies microorganisms in food for safety.
- Methods:
    - Culture methods: Grow microorganisms on media to quantify them.
    - PCR: Detects and identifies specific pathogens quickly.

Q. Mention the physical, chemical, and biological hazardous material which contaminate the food – 10 M

Answer:

Physical hazards
Physical hazards are foreign objects that accidentally enter food during handling or processing.

Examples:

- Glass fragments, metal shavings, plastic pieces, wood splinters
- Bone fragments, stones or pebbles
- Hair, jewelry, dirt, feathers
- Staples, rubber bands, nails, screws
- Insects or insect parts

Chemical hazards
Chemical hazards arise when food comes in contact with harmful chemicals or toxic substances.

Examples:

- Pesticides, herbicides, fungicides
- Veterinary drug residues and excessive food additives
- Cleaning agents such as detergents and sanitizers
- Heavy metals like lead and mercury
- Mycotoxins produced by molds
- Plasticizers, lubricants, industrial contaminants, residual solvents
- Natural plant toxins such as solanine

Biological hazards
Biological hazards are caused by microorganisms or biological substances that contaminate food.

Bacteria: Salmonella, Escherichia coli, Listeria monocytogenes, Campylobacter, Clostridium perfringens

Viruses: Norovirus, Hepatitis A, Rotavirus

Parasites: Giardia lamblia, Toxoplasma gondii, Trichinella spiralis

Fungi: Molds and yeasts

Prions: Associated with mad cow disease

Allergens: Peanuts, tree nuts, milk, eggs, fish, shellfish, soy, wheat and similar substances that trigger allergic reactions.

Q. List out various methods to eliminate the chemical, biological and physical hazardous materials from food – 10 M

Answer:

Elimination of physical hazards
Methods to prevent entry of foreign objects into food:

- Maintain cleanliness and personal hygiene
- Use hair nets, masks, and gloves
- Avoid wearing jewelry while handling food
- Proper maintenance of equipment, buildings, and utilities
- Use magnetic separators to remove metallic particles
- Wash fruits and vegetables thoroughly

Elimination of chemical hazards
Methods to prevent contamination by toxic chemicals:

- Cleaning, washing, and sanitizing surfaces and utensils
- Check quality of water before use
- Follow GAP (Good Agricultural Practices)
- Label and store chemicals separately from food
- Ensure proper ventilation
- Use air curtains to reduce contamination

Elimination of biological hazards
Methods to control microorganisms, insects, and rodents:

- Pest control measures such as fly killers and rat baits
- Maintain proper temperature to prevent microbial growth
- Maintain low moisture, acidity, and oxygen levels in food

Food storage time guidelines:

- Less than 2 hours: may be stored longer
- 2 to 4 hours: cooked, consumed, or discarded
- More than 4 hours: must be discarded

Q. Mention and briefly explain the five methods used for food analysis applicable to samples – 5 M

Answer:

1. Mass spectrometry (MS)

- Used to identify and quantify chemical components in food samples.
- Often combined with liquid chromatography (LC) or gas chromatography (GC) to improve results.
- Can also be coupled with capillary electrophoresis (CE) and infrared spectroscopy (IR).

2. Nuclear magnetic resonance (NMR) spectroscopy

- Uses magnetic properties of atomic nuclei such as hydrogen, carbon-13, phosphorus-31, and deuterium.
- Suitable for analyzing liquid and solid samples.
- Non-destructive method that can detect and quantify multiple compounds simultaneously.
- Known for high reproducibility and versatility.

3. Polymerase chain reaction (PCR)

- Analyzes specific DNA sequences in food samples.
- Used to identify animal species in meat products and prevent food fraud.
- Detects harmful pathogens, microorganisms, and genetically modified organisms (GMO).

4. Enzyme-linked immunosorbent assay (ELISA)

- Detects allergenic proteins, hormones, antibodies, and peptides.
- Highly sensitive method used to detect peanut traces and pathogens such as E. coli, Salmonella, and Listeria.
- Can detect mold spores before growth begins.

5. Lateral flow device (LFD)

- Rapid and efficient technique used for on-site testing.
- Detects substances such as gluten using antigen–antibody reactions.
- Compact, simple, and easy to use.

Q. Expand the term HACCP and explain in detail the 7 principles of HACCP – 10 M

Answer:

HACCP – Expansion

- HACCP stands for Hazard Analysis Critical Control Point.
- It is a management system in which food safety is addressed through the analysis and control of biological, chemical, and physical hazards.
- Covers stages from raw material production, procurement and handling to manufacturing, distribution, and consumption of the finished product.
- The goal of HACCP is to prevent and reduce the occurrence of food safety hazards.

Seven principles of HACCP

1. Conduct a hazard analysis

- List the steps in the process and identify where significant hazards are likely to occur.
- Focus on hazards that can be prevented, eliminated, or controlled.
- Justify inclusion or exclusion of hazards and identify possible control measures.

2. Determine the critical control points (CCPs)

- A CCP is a point, step, or procedure at which control can be applied.
- Control at this stage can prevent, eliminate, or reduce hazards to acceptable levels.
- A CCP decision tree is used to identify CCPs.
- The number of CCPs depends on processing steps and required controls.

3. Establish critical limits

- A critical limit is the maximum or minimum value to which a parameter must be controlled at a CCP.
- Used to prevent, eliminate, or reduce hazards to acceptable levels.
- Critical limits may include time, temperature, pH, water activity, weight, or other measurable parameters.

4. Establish a system to monitor control of the CCP

- Monitoring procedures measure whether the CCP remains within critical limits.

- Procedures specify:
    - How the measurement will be taken
    - When the measurement will be taken
    - Who is responsible
    - How frequently monitoring is performed

5. Establish corrective actions

- Procedures followed when a deviation from a critical limit occurs.
- Steps are taken to prevent hazardous food from entering the food chain.
- Includes identifying the problem and ensuring it does not occur again.

6. Establish verification procedures

- Activities that confirm the HACCP system is working according to the plan.
- May include auditing CCPs, reviewing records, instrument calibration, prior shipment review, and product testing.

7. Establish record-keeping and documentation procedures

- Recording information to prove food was produced safely.
- Records include:
    - HACCP team details
    - Product description
    - Flow diagrams
    - Hazard analysis
    - CCPs and critical limits
    - Monitoring systems
    - Corrective actions
    - Verification procedures and records.

Q. What is Food Hygiene? What are the possible pathogens that can contaminate food and how do you prevent food contamination? – 8 M

Answer:

Food Hygiene

- Food hygiene is the practice of handling, preparing, and storing food in a way that prevents foodborne illnesses.
- It ensures safety and quality of food during preparation, handling, storage, and consumption.

Possible pathogens that can contaminate food

1. Bacteria
    - Salmonella
    - Escherichia coli (E. coli)
    - Listeria monocytogenes
    - Campylobacter

    - Clostridium perfringens
2. Viruses
    - Norovirus
    - Hepatitis A
    - Rotavirus
3. Parasites
    - Giardia lamblia
    - Toxoplasma gondii
    - Trichinella spiralis
4. Fungi
    - Molds
    - Yeasts
5. Prions
    - Associated with diseases such as mad cow disease (BSE).

Prevention of food contamination

1. Proper hygiene
    - Wash hands thoroughly before handling food and after handling raw meat.
2. Safe food storage
    - Store food at correct temperatures.
    - Keep raw foods separate from ready-to-eat foods to prevent cross-contamination.
3. Cooking thoroughly
    - Cook food to recommended internal temperatures to kill pathogens.
4. Regular cleaning
    - Clean kitchen surfaces, utensils, and cutting boards regularly.
    - Sanitize surfaces after preparing raw meat or poultry.
5. Food sourcing
    - Purchase food from reliable suppliers.
    - Check expiration dates and avoid questionable sources.
6. Temperature control
    - Use thermometers to monitor cooking and storage temperatures.
    - Avoid leaving food at room temperature for more than 2 hours.
7. Safe thawing
    - Thaw food in a refrigerator, cold water, or microwave, not at room temperature.
8. Educating staff
    - Train food handlers in safe handling practices and hygiene.
9. Monitoring and labeling
    - Regular inspection of storage and preparation areas.
    - Proper labeling with preparation and expiry dates to reduce spoilage.

Q. Elaborate on the various instruments required for Food Analysis and Testing in laboratories for maintaining the food quality – 8 M

Answer:

Instruments used in food analysis and testing laboratories

1. **Mass Spectrometer (MS)**
    - Used to identify and quantify chemical components in food samples.
    - Often coupled with gas chromatography (GC) or liquid chromatography (LC) to improve accuracy.
    - Useful for detecting contaminants, additives, and chemical composition.
2. **Nuclear Magnetic Resonance (NMR) Spectrometer**
    - Uses magnetic properties of atomic nuclei to analyze food samples.
    - Non-destructive method suitable for both liquid and solid foods.
    - Can detect and quantify multiple compounds simultaneously.
3. **PCR (Polymerase Chain Reaction) Instrument**
    - Used to analyze specific DNA sequences.
    - Identifies pathogens, microorganisms, and genetically modified organisms (GMOs).
    - Also useful in species identification in meat products.
4. **ELISA Reader (Enzyme-Linked Immunosorbent Assay)**
    - Detects allergenic proteins, hormones, antibodies, and peptides.
    - Highly sensitive method for detecting pathogens and allergens.
    - Used in dairy and processed food testing.
5. **Lateral Flow Device (LFD) Kits**
    - Rapid on-site testing method.
    - Detects substances such as gluten using antigen–antibody reactions.
    - Compact and easy to use.
6. **Chromatography Instruments (GC / HPLC)**
    - Gas Chromatography (GC): Separates and identifies volatile compounds and contaminants.
    - High-Performance Liquid Chromatography (HPLC): Used to analyze sugars, vitamins, and other non-volatile compounds.
7. **Spectrophotometer**
    - Measures absorbance of light to determine concentration of compounds such as vitamins and enzymes.
    - Used in biochemical and nutrient analysis.
8. **pH Meter**
    - Measures acidity or alkalinity of food.
    - Important for assessing food safety, preservation, and quality.

These instruments help maintain food quality by analyzing composition, detecting contaminants, verifying nutritional value, and ensuring compliance with food safety standards.

Q. Analyze the implementation challenges of food safety standards in the Indian unorganized food sector. Suggest practical solutions – 8 M

Answer:

Implementation challenges in the unorganized food sector

1. Lack of licensing and regulatory compliance: Food licenses are compulsory for food business operators, but many small vendors and petty businesses may fail to obtain licenses or follow labeling and safety standards.

2. Poor hygiene and sanitation practices: Food safety compliance requires proper hygiene and sanitation in processing and handling, which is often difficult to maintain in small or informal setups.

3. Limited awareness and training: Food safety systems require knowledge of standards, hygiene, and handling practices, and lack of training among small vendors creates difficulty in implementation.

4. Inadequate infrastructure and facilities: Safe food production requires proper storage, temperature control, and handling conditions, which may not be available in small-scale operations.

5. Irregular food testing and quality control: Regular food testing and laboratory analysis are necessary to ensure quality and safety, but small businesses often lack access to testing facilities.

6. Difficulty in maintaining records and documentation: Food safety systems require labeling, documentation, and record keeping, which may be difficult for informal vendors to maintain.

Practical solutions

1. Awareness and training programmes: Conduct training programmes to educate food handlers and small vendors on hygiene, safety standards, and proper handling practices.

2. Simplified licensing and support for small businesses: Basic registration systems for small businesses allow gradual compliance as operations grow.

3. Implementation of Good Hygiene Practices (GHP) and Good Manufacturing Practices (GMP): These practices help prevent contamination and ensure safe production, handling, and storage of food.

4. Improved inspection and monitoring: Regular inspections, audits, and testing help ensure compliance and protect consumers.

5. Access to food testing and quality control facilities: Laboratory testing ensures food safety, detects contaminants, and maintains quality standards.

6. Better hygiene and sanitation practices: Maintaining clean environments, proper storage, and safe handling reduces contamination and foodborne illnesses.

Q. Discuss and elaborate on techniques used in the food testing laboratories – 8 M

Answer:

Techniques used in food testing laboratories

1. Microbiological testing:
Purpose: To detect and identify microorganisms such as bacteria, viruses, and fungi that can cause foodborne illnesses.
Methods: Culture-based procedures to grow and count microorganisms, PCR to detect pathogen DNA, and immunoassays to detect specific antigens.

2. Chemical testing:
Purpose: To analyze chemical composition, contaminants, and residues in food.
Methods: Gas chromatography (GC) and mass spectrometry (MS) for identifying compounds and contaminants, HPLC for analyzing non-volatile compounds, spectroscopy techniques such as IR and NMR, and residue analysis for pesticides and heavy metals.

3. Physical testing:
Purpose: To evaluate physical properties affecting quality and appearance.
Methods: Visual inspection for color and appearance, texture analysis for hardness or chewiness, and density or viscosity measurements.

4. Sensory evaluation:
Purpose: To assess food quality based on taste, smell, sight, and texture.
Methods: Panel testing by trained evaluators and consumer testing for product feedback.

5. Allergen testing:
Purpose: To detect common allergens such as peanuts, gluten, soy, and dairy.
Methods: ELISA to detect allergen traces and PCR to identify allergenic DNA sequences.

6. Shelf life testing:
Purpose: To determine how long food remains safe and acceptable under storage conditions.
Methods: Accelerated shelf life testing, microbial growth analysis, and chemical deterioration testing such as oxidation.

Q. Discuss the causes, symptoms, and prevention of foodborne illnesses – 6 M

Answer:

Causes of foodborne illnesses

1. Microorganisms: Foodborne illnesses may result from bacteria, viruses, fungi, and parasites present in contaminated food, raw or undercooked meat, unpasteurized dairy products, contaminated fruits and vegetables, or drinking water.

2. Prions and naturally occurring toxins: Infectious proteins such as those associated with mad cow disease and naturally occurring toxins in foods such as mushrooms, cereals, or mold on grains can cause illness.

3. Environmental contamination: Environmental pollutants, heavy metals, and contaminants present in soil or water can enter the food chain and lead to foodborne illness.

4. Improper handling and storage: Food spoilage and microbial growth occur due to improper temperature, moisture, or storage conditions, as well as insect or physical damage.

Symptoms of foodborne illnesses

1. Diarrhea
2. Vomiting
3. Abdominal cramps
4. Fever
5. Nausea
6. General weakness and dehydration

Symptoms vary depending on the microorganism and severity of contamination.

Prevention of foodborne illnesses

1. Maintain proper food hygiene: Handle, prepare, and store food in a clean and hygienic manner to prevent contamination.

2. Proper cooking and storage: Cook food thoroughly and store food at appropriate temperatures to prevent microbial growth.

3. Safe water and raw materials: Use clean water and ensure raw materials are free from contamination.

4. Prevent cross-contamination: Keep raw foods separate from cooked or ready-to-eat foods and maintain cleanliness of utensils and surfaces.

5. Control of environmental and microbial contamination: Proper sanitation, pest control, and safe handling practices help reduce contamination and spoilage.

Q. Identify and elaborate on the type of hazard in each step of the wheat processing – 7 M

Answer:

Steps in wheat processing

- Farming and harvesting
- Storage in grain elevators
- Cleaning, separation, and transportation
- Flour milling and processing
- Baking / preparation and consumption

1. Farming and harvesting stage

- Biological hazards: Microorganisms present in soil, water, or on plants; mold growth on grains due to moisture.
- Chemical hazards: Pesticide, herbicide, and fungicide residues used during cultivation.

- Physical hazards: Stones, soil particles, plant debris, and other foreign materials mixed with harvested grain.

2. Storage in grain elevators

- Biological hazards: Growth of fungi and bacteria due to improper storage; insect infestation and pest contamination.
- Chemical hazards: Fumigants or storage chemicals used for pest control.
- Physical hazards: Dust, husk particles, or foreign objects entering grain during storage.

3. Cleaning, separation, and transportation

- Biological hazards: Contamination due to poor hygiene or handling during transport.
- Chemical hazards: Lubricants, oils, or cleaning chemicals from machinery and transport equipment.
- Physical hazards: Metal fragments, plastic pieces, or packaging materials entering during handling and transport.

4. Flour milling and processing

- Biological hazards: Microbial contamination from equipment or environment.
- Chemical hazards: Residues from cleaning agents or additives used in processing.
- Physical hazards: Metal particles from machinery, dust, or broken equipment parts.

5. Baking, preparation, and consumption stage

- Biological hazards: Cross-contamination from food handlers, utensils, or environment.
- Chemical hazards: Contamination from additives or packaging materials.
- Physical hazards: Foreign materials such as hair, plastic pieces, or packaging fragments introduced during handling or preparation.

Q. Mention and justify the elimination method for identified hazard during the wheat processing – 7 M

Answer:

Elimination methods during wheat processing

1. Farming and harvesting stage

- Elimination of biological hazards:
  Use GAP, control moisture, monitor plant health.
  Justification: Prevents growth of microorganisms and fungi.
- Elimination of chemical hazards:
  Use pesticides in controlled quantities.
  Justification: Reduces chemical residues on grains.

- Elimination of physical hazards:
  Cleaning and sorting of harvested grain.
  Justification: Removes stones, soil, and debris.

2. Storage in grain elevators

- Elimination of biological hazards:
  Maintain proper temperature, humidity, and pest control.
  Justification: Prevents microbial growth and infestation.
- Elimination of chemical hazards:
  Controlled use of fumigants and storage chemicals.
  Justification: Prevents toxic residues.
- Elimination of physical hazards:
  Cleaning of storage units and proper handling.
  Justification: Prevents entry of dust and foreign objects.

3. Cleaning, separation, and transportation

- Elimination of biological hazards:
  Maintain hygiene of equipment and vehicles.
  Justification: Prevents contamination during handling.
- Elimination of chemical hazards:
  Proper cleaning of equipment and safe storage of chemicals.
  Justification: Avoids chemical contamination.
- Elimination of physical hazards:
  Use sieving, screening, and magnetic separators.
  Justification: Removes metal fragments and particles.

4. Flour milling and processing

- Elimination of biological hazards:
  Regular cleaning and sanitation of equipment.
  Justification: Reduces microbial contamination.
- Elimination of chemical hazards:
  Use food-grade additives and controlled cleaning agents.
  Justification: Prevents chemical residues.
- Elimination of physical hazards:
  Equipment maintenance and filtration.
  Justification: Prevents entry of metal particles.

5. Baking, preparation, and consumption stage

- Elimination of biological hazards:
  Proper cooking and personal hygiene.
  Justification: Heat destroys pathogens and prevents contamination.
- Elimination of chemical hazards:
  Use approved additives and safe packaging.
  Justification: Ensures food safety.

- Elimination of physical hazards:
  Use hair nets, gloves, and proper handling.
  Justification: Prevents foreign materials entering food.

Q. Elaborate on food allergens – 7 M

Answer:

Food allergens

Food allergens are substances, typically proteins in food, that can trigger an allergic reaction in susceptible individuals.
The immune system mistakenly identifies the allergen as harmful and releases chemicals such as histamine.

How food allergies occur

1. Consumption of food containing the allergen.
2. Immune system overreacts and releases chemicals.
3. Allergic symptoms are produced.

Symptoms of food allergy

Mild symptoms

- Hives
- Itching
- Swelling of mouth, lips, or tongue

Severe symptoms

- Difficulty breathing
- Wheezing
- Dizziness or fainting
- Anaphylaxis (life-threatening reaction)

Major food allergens

- Milk
- Eggs
- Fish
- Shellfish (crustaceans)
- Tree nuts (almonds, walnuts, cashews, hazelnuts)
- Peanuts
- Wheat
- Soybeans
- Sesame

Other allergens may include celery, mustard, and certain fruits or vegetables.

Important points about food allergens

1. Food intolerance is not the same as food allergy:
   Food intolerance does not involve the immune system.
2. Any food can cause a reaction:
   Most allergies are triggered by a limited number of foods.
3. Cross-contamination is a concern:
   Even small amounts of an allergen can trigger a reaction.
4. Accurate labeling is essential:
   Food labels must clearly identify the presence of major allergens.

Allergen testing

- Helps detect target allergens in ingredients and finished food products.
- Commonly tested allergens include peanuts, gluten, eggs, nuts, and milk.
- Methods used:
    - ELISA to detect traces of allergens
    - PCR to identify allergenic DNA sequences